



Welcome to CSC 365

Internet Programming

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What are CSS Libraries?

Definition:

Software libraries that abstract away the low-level CSS implementation of user-facing elements.

Some popular libraries include...

- Bootstrap
- Foundation
- Semantic UI
- Pure
- UIKit

Bootstrap

- <https://getbootstrap.com/>
- Many, many, many (new) websites use Bootstrap!
- get (oneself or something) into or out of a situation using existing resources.



How Bootstrap Works

Bootstrap provides us with...

- Layouts
- Content
- Components
- Utilities

There is much more!

Bootstrap Categories:

Containers are the most basic element of layouts.

```
<div class="container">  
  ...  
</div>
```

```
<div class="container-fluid">  
  ...  
</div>
```

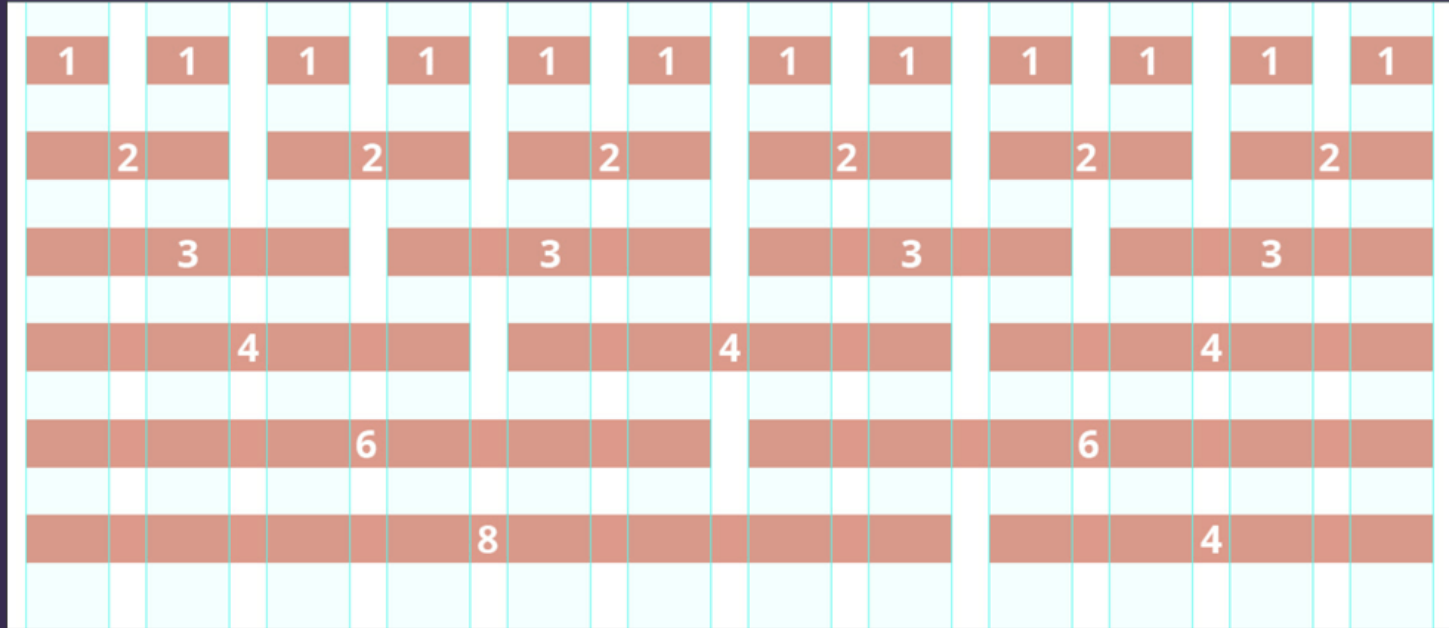
Grids

Often, containers will contain **rows** and **columns** to form a grid...

```
<div class="container">  
  <div class="row">  
    <div class="col-*"></div>  
    <div class="col-*"></div>  
  </div>  
</div>
```

Where * is the column span.

Grid System



Responsive Design

Definition:

Responsive design adapts content to a variety of devices and screen sizes.

Width break points determine whether the design will scale or be reorganized.

Responsive Design

What if we want our webpage to respond to different screen sizes, e.g. phone, tablet, and monitor?

```
<div class="container">  
  <div class="row">  
    <div class="col-12 col-md-6 col-lg-3"></div>  
    <div class="col-12 col-md-6 col-lg-3"></div>  
    <div class="col-12 col-md-6 col-lg-3"></div>  
    <div class="col-12 col-md-6 col-lg-3"></div>  
  </div>  
</div>
```

example

```
12 <body>
13   <h1 class="test">News 1</h1>
14   <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5/dist/css/bootstrap.min.css">
15   <div class="container py-4">
16     <div class="row g-3">
17       <div class="col-md-8">
18         <div class="card">
19           <div class="card-body">
20             <h5 class="card-title">Hello, Bootstrap</h5>
21             <p class="card-text">Quickly build responsive layouts.</p>
22             <a class="btn btn-primary" href="#">Get started</a>
23           </div>
24         </div>
25       </div>
26       <div class="col-md-4">
27         <div class="alert alert-info mb-0">Responsive by default!</div>
28       </div>
29     </div>
30   </div>
31
32 </body>
```

example

News 1

Hello, Bootstrap

Quickly build responsive layouts.

Get started

Responsive by default!

News 1

Hello, Bootstrap

Quickly build responsive layouts.

Get started

Responsive by default!

Bootstrap Categories: Content

Content styling includes basic HTML elements, typography, code, images, tables, figures.

Basic HTML examples:

```
<h1></h1>  
<ul></ul>  
<input/>  
<button></button>
```

These will get the default Bootstrap styling.

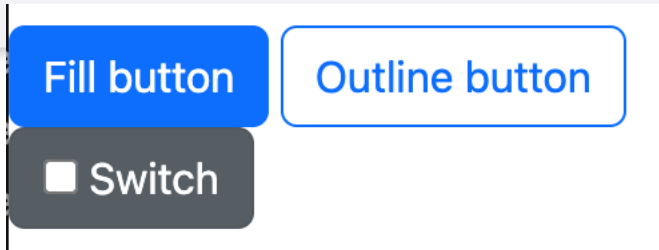
Bootstrap Categories: Components

Components include all other visual/interactive elements that make up the design, e.g., buttons, forms, navbar, tooltips, etc.

```
<button type="button" class="btn btn-primary">Fill button</button>

<button type="button" class="btn btn-outline-primary">Outline button</button>

<div class="btn-group-toggle" data-toggle="buttons">
  <label class="btn btn-secondary active">
    <input type="checkbox" checked autocomplete="off"> Switch
  </label>
</div>
```



Bootstrap Categories: Utilities

Utilities are not elements themselves, but they modify/control other elements, e.g., adding rounded corners to an image.

```

```

```
<div class="shadow p-3 mb-5 bg-white rounded">Shadow</div>
```

Responsive by default!